

1. Create and Deploy Blockchain Network using Hyperledger Fabric SDK for Java

Procedure

1. Install required tools like Docker, Java, Maven, Git and Node.js.
2. Start the Docker service.
3. Clone the Fabric Java SDK project from GitHub.
4. Open the network folder.
5. Stop and remove old network files.
6. Build the Fabric network.
7. Open the Java application folder.
8. Build the Java client using Maven.
9. Create and initialize the channel.
10. Deploy chaincode and perform invoke and query operations.

Important Commands

```
sudo apt update
```

```
sudo apt install docker.io git maven nodejs npm
```

```
git clone https://github.com/IBM/blockchain-application-using-fabric-java-sdk.git
```

```
cd blockchain-application-using-fabric-java-sdk/network
```

```
./build.sh
```

```
cd ../java
```

```
mvn install
```

```
java -cp blockchain-client.jar org.example.network.CreateChannel
```

```
java -cp blockchain-client.jar org.example.chaincode.invocation.QueryChaincode
```

2. Develop a Blockchain App to Execute Transactions and Requests

Procedure

1. Install Node.js and npm.
2. Install Ganache for local blockchain.
3. Create a new project folder.
4. Initialize the Node.js project.
5. Install Web3 package.
6. Create a JavaScript file for transaction.
7. Connect the app with Ganache blockchain.
8. Get the blockchain accounts.
9. Transfer amount from one account to another.
10. Display transaction hash and balance details.

Important Commands

```
sudo apt update
```

```
sudo apt install nodejs npm
```

```
npm install -g ganache
```

```
mkdir transaction-app
```

```
cd transaction-app
```

```
npm init -y
```

```
npm install web3
```

```
ganache
```

```
node app.js
```

3. Develop an Asset-Transfer App using Blockchain

Procedure

1. Install Docker, Go, Node.js and npm.
2. Download Hyperledger Fabric samples.
3. Open the Fabric test network folder.
4. Stop the old network.
5. Start a new Fabric network.
6. Create a channel.
7. Deploy asset-transfer chaincode.
8. Initialize the ledger with sample assets.
9. Query all assets from the ledger.
10. Run the JavaScript asset-transfer application.

Important Commands

```
sudo apt update
```

```
sudo apt install docker.io nodejs npm golang-go
```

```
curl -sSLO https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh
```

```
chmod +x install-fabric.sh
```

```
./install-fabric.sh docker samples binary
```

```
cd fabric-samples/test-network
```

```
./network.sh down
```

```
./network.sh up createChannel -ca
```

```
./network.sh deployCC -ccn basic -ccp ../asset-transfer-basic/chaincode-go -ccl go
```

```
node app.js
```

4. Develop a Blockchain to Track Fitness Club Rewards

Procedure

1. Install Docker, Go, Node.js and Fabric tools.
2. Create a folder for fitness reward chaincode.
3. Create Go module for the chaincode.
4. Write chaincode for member reward details.
5. Add function to issue reward points.
6. Add function to view reward points.
7. Start the Fabric test network.
8. Deploy fitness reward chaincode.
9. Invoke reward issue transaction.
10. Query the reward details from blockchain.

Important Commands

```
sudo apt update
```

```
sudo apt install docker.io nodejs npm golang-go
```

```
mkdir fitness-rewards-chaincode
```

```
cd fitness-rewards-chaincode
```

```
go mod init fitnessrewards
```

```
go mod tidy
```

```
cd ../fabric-samples/test-network
```

```
./network.sh up createChannel -ca
```

```
./network.sh deployCC -ccn rewards -ccp ../fitness-rewards-chaincode -ccl go
```

```
peer chaincode query -C mychannel -n rewards
```

5. Develop a Car Auction Network using Hyperledger Fabric Node SDK

Procedure

1. Install Docker, Go, Node.js and Fabric.
2. Create a folder for car auction chaincode.
3. Create Go module for the auction chaincode.
4. Write chaincode to create auction.
5. Add function to place bid.
6. Add function to view auction result.
7. Start Hyperledger Fabric test network.
8. Deploy car auction chaincode.
9. Create a car auction record.
10. Place bid and query auction details.

Important Commands

```
sudo apt update
```

```
sudo apt install docker.io nodejs npm golang-go
```

```
mkdir car-auction-chaincode
```

```
cd car-auction-chaincode
```

```
go mod init carauction
```

```
go mod tidy
```

```
cd ../fabric-samples/test-network
```

```
./network.sh up createChannel -ca
```

```
./network.sh deployCC -ccn carauction -ccp ../car-auction-chaincode -ccl go
```

```
peer chaincode query -C mychannel -n carauction
```

6. Develop a Blockchain Based App to Manage Healthcare System

Procedure

1. Install Docker, Go and Hyperledger Fabric.
2. Create a healthcare chaincode folder.
3. Create Go module for the healthcare app.
4. Write patient structure in chaincode.
5. Add function to store patient record.
6. Add function to view patient details.
7. Start Fabric test network.
8. Deploy healthcare chaincode.
9. Add one patient record.
10. Query and verify the patient record.

Important Commands

```
sudo apt update
```

```
sudo apt install docker.io golang-go nodejs npm
```

```
mkdir healthcare-chaincode
```

```
cd healthcare-chaincode
```

```
go mod init healthcare
```

```
go mod tidy
```

```
cd ../fabric-samples/test-network
```

```
./network.sh up createChannel -ca
```

```
./network.sh deployCC -ccn healthcare -ccp ../healthcare-chaincode -ccl go
```

```
peer chaincode query -C mychannel -n healthcare
```

7. Implement Blockchain Based Supply Chain Transaction System

Procedure

1. Install Docker, Go and Fabric tools.
2. Create supply chain chaincode folder.
3. Create Go module for supply chain system.
4. Write chaincode to store product transaction.
5. Add function to record supply chain transaction.
6. Add function to view transaction details.
7. Start the Fabric test network.
8. Deploy supply chain chaincode.
9. Add one supply chain transaction.
10. Query and verify the transaction details.

Important Commands

```
sudo apt update
```

```
sudo apt install docker.io golang-go nodejs npm
```

```
mkdir supply-chain-chaincode
```

```
cd supply-chain-chaincode
```

```
go mod init supplychain
```

```
go mod tidy
```

```
cd ../fabric-samples/test-network
```

```
./network.sh up createChannel -ca
```

```
./network.sh deployCC -ccn supplychain -ccp ../supply-chain-chaincode -ccl go
```

```
peer chaincode query -C mychannel -n supplychain
```

8. Create Blockchain Based Online Voting System

Procedure

1. Install Docker, Go and Hyperledger Fabric.
2. Create online voting chaincode folder.
3. Create Go module for voting system.
4. Write candidate structure in chaincode.
5. Add function to add candidate.
6. Add function to vote for candidate.
7. Add function to view vote count.
8. Start Fabric test network.
9. Deploy voting chaincode.
10. Add candidate, vote and query result.

Important Commands

```
sudo apt update
```

```
sudo apt install docker.io golang-go nodejs npm
```

```
mkdir voting-chaincode
```

```
cd voting-chaincode
```

```
go mod init voting
```

```
go mod tidy
```

```
cd ../fabric-samples/test-network
```

```
./network.sh up createChannel -ca
```

```
./network.sh deployCC -ccn voting -ccp ../voting-chaincode -ccl go
```

```
peer chaincode query -C mychannel -n voting
```


9. Develop Blockchain Based Password Manager

Procedure

1. Install Docker, Go and Fabric tools.
2. Create password manager chaincode folder.
3. Create Go module for password manager.
4. Write password record structure.
5. Store service name, username and password hash.
6. Add function to store password record.
7. Add function to view password record.
8. Start Fabric test network.
9. Deploy password manager chaincode.
10. Store and query password record.

Important Commands

```
sudo apt update
```

```
sudo apt install docker.io golang-go nodejs npm
```

```
mkdir password-manager-chaincode
```

```
cd password-manager-chaincode
```

```
go mod init passwordmanager
```

```
go mod tidy
```

```
cd ../fabric-samples/test-network
```

```
./network.sh up createChannel -ca
```

```
./network.sh deployCC -ccn passwordmanager -ccp ../password-manager-chaincode -ccl go
```

```
peer chaincode query -C mychannel -n passwordmanager
```

10. Develop Blockchain Based Donation System

Procedure

1. Install Docker, Go and Fabric tools.
2. Create donation chaincode folder.
3. Create Go module for donation system.
4. Write donation structure in chaincode.
5. Add donor, receiver, amount and purpose fields.
6. Add function to store donation details.
7. Add function to view donation details.
8. Start Hyperledger Fabric test network.
9. Deploy donation chaincode.
10. Add donation record and query the output.

Important Commands

```
sudo apt update
```

```
sudo apt install docker.io golang-go nodejs npm
```

```
mkdir donation-chaincode
```

```
cd donation-chaincode
```

```
go mod init donation
```

```
go mod tidy
```

```
cd ../fabric-samples/test-network
```

```
./network.sh up createChannel -ca
```

```
./network.sh deployCC -ccn donation -ccp ../donation-chaincode -ccl go
```

```
peer chaincode query -C mychannel -n donation
```